

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460568.8358 +/- 0.003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.203 +/- 0.018
Transit depth (Rp/Rs)^2	4.13 +/- 0.74 [%]
Orbital Inclination (inc)	85.28 +/- 1.27
Airmass coefficient 1 (a1)	1.056 +/- 0.025
Airmass coefficient 2 (a2)	-0.044 +/- 0.038
Scatter in the residuals of the lightcurve fit is	2.37 %
Best Comparison Star	#2 - [179, 116]
Optimal Aperture	3.66
Optimal Annulus	7.98
Transit Duration (day)	0.078 +/- 0.0074

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.203 $\pm$ 0.018 vs 0.1646 (deviation 2.13 σ)
Duration fit vs published	0.078 d vs 0.0758 d (deviation 0.29 σ)
Mid-time O-C	2.2 min
Depth SNR	11.3
Residual scatter	2.37 %

Light curve

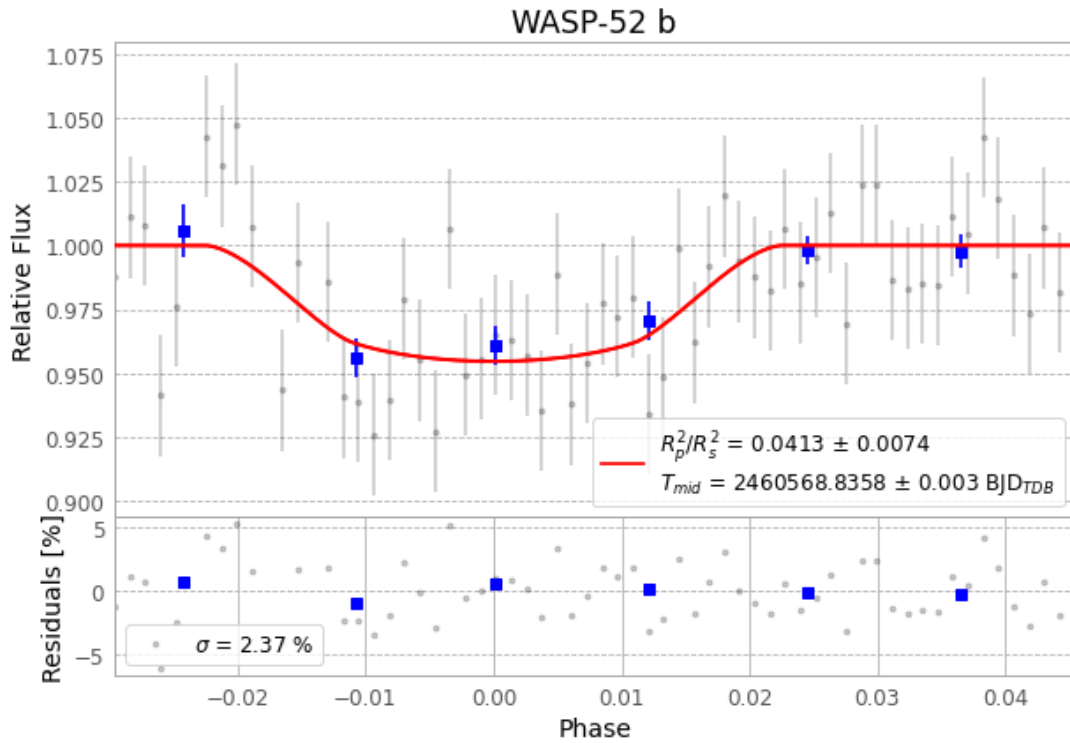


Figure 1 — FinalLightCurve_WASP-52 b_2024-09-14.png (SHA-256 aa9c49d38b5a9dc7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [299, 239], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1dfcf3e0c9397117...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

64 FITS frames (42 MB), WASP-52240915063916.FITS → WASP-52240915094815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-240915.log (f3ea681aba48...), FinalLightCurve_WASP-52 b_2024-09-14.png (aa9c49d38b5a...), FinalLightCurve_WASP-52 b_2024-09-14.csv (eb012e3d6f1c...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-21 23:29Z.